M Reproducibility Checklist for JAIR

This appendix answers the JAIR reproducibility checklist (Gundersen et al. 2024) for the submission version of this article, as required for review.

All articles:

(1) All claims investigated in this work are clearly stated. **Yes.**
(2) Clear explanations are given how the work reported substantiates the claims. **Yes.** Proofs accompany the statements or appear in Appendices A and H. Empirical claims are supported by committed result files and their reported summaries.
(3) Limitations or technical assumptions are stated clearly and explicitly. **Yes.** The statement assumptions, Example H.2, discussion and conclusion delimit the structural, reward-convention and empirical claims.
(4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. **Yes.** The recursive EFE agent (Equation 3), the usage-curve procedure (Section 4), and the online dual controller update are given in closed form in the main text, with build and export details for the SARSOP baseline in Appendix G.1.
(5) Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics. **Yes.** Section 3 motivates the observe-then-commit restriction, and Section 4.6 positions every budgeted-formulation design choice against prior art.

Articles containing theoretical contributions:

Does this paper make theoretical contributions? **Yes.**

(1) All assumptions and restrictions are stated clearly and formally. **Partially.** The definitions and propositions state their structural and regularity assumptions. The information base, bits in the implementation and nats in the canonical identification, is specified in the surrounding text rather than inside Proposition 3.1.
(2) All novel claims are stated formally (e.g., in theorem statements). **Yes.** Propositions 3.1, H.1, 3.3, 4.2, 4.3, 4.5, Definition 4.1, and Corollary 4.4.
(3) Proofs of all non-trivial claims are provided in sufficient detail to permit verification by readers with a reasonable degree of expertise (e.g., that expected from a PhD candidate in the same area of AI). **Partially.** Appendix A proves Proposition 3.1. The remaining statements have proofs or case-analysis sketches at their statements or in Appendix H. The supermartingale and running-average proof covers only the projected decaying-step controller, not the released constant-step default or reset-on-shift heuristic.
(4) Complex formalism, such as definitions or proofs, is motivated and explained clearly. **Yes.**
(5) The use of mathematical notation and formalism serves the purpose of enhancing clarity and precision; gratuitous use of mathematical formalism (i.e., use that does not enhance clarity or precision) is avoided. **Yes.**
(6) Appropriate citations are given for all non-trivial theoretical tools and techniques. **Yes.** Log scoring (Bernardo 1979), the Robbins–Monro recursion (Robbins and Monro 1951), classical almost-sure extensions (Blum 1954), background on constrained stochastic approximation (Kushner and Clark 1978), and the sophisticated-inference Bellman-optimality result (Da Costa, Sajid, et al. 2023) that the recursive agent is built on.

Articles reporting on computational experiments:

Does this paper include computational experiments? **Yes.**

(1) All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for source

code readability and documentation as well as for long-term accessibility. **Partially.** All experiment code, installation instructions and reproduction commands are public at https://github.com/PatrickAllenCooper/rho_aif. The repository includes tests, pinned dependencies and documented result schemas. Long-term preservation beyond the versioned GitHub repository is not established. PyPI distribution as `rho-aif` is planned but not yet published.

(2) The source code comes with a license that allows free usage for reproducibility purposes. **Yes.** MIT License.

(3) The source code comes with a license that allows free usage for research purposes in general. **Yes.** MIT License.

(4) Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility. **Partially.** Summary CSVs support every reported simulation result, but episode-level records are incomplete. Per-episode dual-control traces, per-seed SARSOP TOST results (including the n=20 and episode-matched checks), budget-frontier and same-stream reference rows, fresh-seed and target-matched gaps, and usage-curve/bracket-bootstrap files are committed. Appendix I.1 additionally archives every sensor-study candidate trajectory, model, split, selection and bootstrap output. UCI supplies the original measurements under a pinned checksum.
Other simulation trajectories can be regenerated using the recorded environment and README command map. Core, Tileworld, Inspection, scaling, discount and navigation summaries carry seed, episode, revision and timestamp provenance. RockSample records seeds and episode counts. Offline-reference and calibration protocols are stated in the text, while the MCTS ablation and POMCP sweep carry full provenance. Relevant summaries include seed-level SEs and Welch/Holm companions. Complete episode-level simulation records are not available for every retained battery. Long-term preservation beyond the versioned repository is not established.

(5) The unaggregated data comes with a license that allows free usage for reproducibility purposes. **Yes.** The repository's MIT License covers its code and generated simulation outputs. The external sensor measurements retain the primary source's license and attribution terms (Vergara 2012).

(6) The unaggregated data comes with a license that allows free usage for research purposes in general. **Yes.** The sensor archive is obtained from UCI under its stated research-use and attribution terms.

(7) If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results. **Yes.** Main simulation results use $\{42, 123, 456, 789, 1024\}$. Weight tuning uses stream 7 (Appendix H.4). The frontier uses held-out seeds 7–11 and fresh seeds 12–21, both at 100 episodes/seed. The n=20 SARSOP checks add seeds 2000–2014. POMCP constants use tuning seeds $\{11, 22, 33\}$ at 100 episodes/seed, with evaluation on the canonical five. Counts for the ten-seed dual study, ten-seed smaller RockSample instances and other retained batteries are stated with their results.
The benchmark, budget, RockSample, Inspection, distractor, frontier and dual runners reset each episode at $10^4 s + i$, for outer seed s and episode index i. SARSOP, CPOMDP, TOST and the observe-then-commit calibration runner instead seed once per outer stream, except for TOST's explicit `-episode-seeding` mode. POMCP's internal seed varies with the outer seed in its main comparison and exploration sweep, while the MCTS-EFE and compute-matched batteries retain a fixed internal seed (Appendix F.2). The dual rescale is also seeded and covered by a trajectory-reproduction regression test.
Sensor-study split/quantizer, mixture-check and bootstrap seeds are 20261001, 20261002 and 20261003, with a derived later-batch stream. Its bootstrap draws are conditional case resamples, not independent model fits. Exact reproduction uses the recorded environment or archived trajectories.

(8) The execution environment for experiments, the computing infrastructure (hardware and software) used for running them, is described, including GPU/CPU makes and models; amount of memory (cache and

RAM); make and version of operating system; names and versions of relevant software libraries and frameworks. **Partially.** All experiments are CPU-only. The simulation environment of record is an Apple M5 Max with 36 GB RAM, macOS 26.6.2 and Python 3.9.6. The recorded-sensor study uses macOS 27.0 on arm64 with Python 3.9.6, as documented in its manifests and Appendix I.1. `ENVIRONMENT.md` records the machine and solver build, `requirements-lock.txt` pins package versions, and `pyproject.toml` gives installation requirements. Cache sizes were not recorded, and the sensor-study manifests do not independently record the processor model or installed RAM.

(9) The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated. **Yes.** Success rate, mean reward, mean observation count, and usage (sensing spend) are defined at first use and motivated by the observe-then-commit setting's cost structure (Section 3).

(10) The number of algorithm runs used to compute each result is reported. **Yes.** Five seeds per condition unless stated otherwise (e.g., ten seeds for the multi-seed dual-control study of Section 6.2). Episode counts are reported per benchmark in the seeding item above and in every results table caption.

(11) Reported results have not been "cherry-picked" by silently ignoring unsuccessful or unsatisfactory experiments. **Yes.** Section 8 and the conclusion explicitly report where the proposed method does not help (Section 6.10's distractor robustness result, Example H.2's destructive-sensing failure mode, and cases where tuned Planning+IG exceeds EFE on particular metrics).

(12) Analysis of results goes beyond single-dimensional summaries of performance (e.g., average, median) to include measures of variation, confidence, or other distributional information. **Yes.** Standard errors and confidence intervals are reported throughout Sections 6 and 7 (e.g., the paired Student-t interval for the recovery-time difference in Section 6.2 and the usage/reward SEs in Table 2).

(13) All (hyper-) parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them. **Yes.** Planning horizons and discount factors are reported per environment (Appendix C) and the tuned Planning+IG weight search protocol is stated in Section 3, with discount-factor sensitivity swept explicitly in Appendix D.2.

(14) The number and range of (hyper-) parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-) parameter optimisation. **Yes.** The Pareto sweep (Section 7.2) reports every searched weight. Appendix F.2 states the exploration grid, selection rule, tuning seeds and retrospective design qualification. Appendix I.1 gives the frozen sensor-study grids.

(15) Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects. **Yes.** Confidence intervals and seed-level t-tests (`tests/test_stats_seed.py`) are used rather than point-estimate comparisons alone. The SARSOP comparison of Section 6.6 is additionally backed by a fixed-margin TOST equivalence test (Schuirmann 1987). The near-optimal CPOMDP reference of Section 6.7 reports the measured reward gap to a sampled cap-feasible reference from a Lagrangian-penalized SARSOP sweep. It is evaluated by simulation and is not a certified optimality bound.

Articles using data sets:

Does this work rely on one or more data sets (possibly obtained from a benchmark generator or similar software artifact)? **Yes.** Section 6.9 uses the public UCI Gas Sensor Array Drift dataset (Vergara 2012; Vergara et al. 2012). The primary source, download checksum, original row identities, deterministic training/calibration/test split and later-batch holdout are archived. Appendix I.1 describes the sensor grouping, training-only transforms, fixed targets and model restrictions. There are no exact duplicate or cross-period overlapping feature vectors. Access costs are benchmark conventions, not measured physical costs. Other environments are simulators defined generatively in code (Appendix C). Our cost-modified RockSample is not the standard generator and its absolute returns are not comparable to published RockSample numbers.

(1) All newly introduced data sets are included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility with a license that allows free usage for research purposes. **Partially.** Generated simulation summaries and the recorded-sensor study's model, splits and per-case policy outputs are public in the repository. Not every simulation trajectory is retained, as detailed in the computational-experiments answers above. Repository code and generated simulation outputs are MIT licensed. The original UCI measurements remain at their primary source under its terms. Long-term preservation beyond the versioned repository is not established.
(2) The newly introduced data set comes with a license that allows free usage for reproducibility purposes. **Yes.** The repository's MIT License permits reuse of its generated simulation outputs for reproduction. The pre-existing UCI measurements retain their source license and attribution requirements (Vergara 2012).
(3) The newly introduced data set comes with a license that allows free usage for research purposes in general. **Yes.** The MIT License permits general research reuse of the repository's generated simulation outputs. This does not replace the source terms for the pre-existing UCI measurements.
(4) All data sets drawn from the literature or other public sources (potentially including authors' own previously published work) are accompanied by appropriate citations. **Yes.** Section 6.9 cites the UCI dataset record and the associated study (Vergara 2012; Vergara et al. 2012). The simulated environments are identified in Appendix C, including the material differences from standard RockSample.
(5) All data sets drawn from the existing literature (potentially including authors' own previously published work) are publicly available. **Yes.** The Gas Sensor Array Drift dataset is publicly available from UCI (Vergara 2012). The repository's sensor-study instructions give the download procedure and pinned source checksum. The original measurements are retrieved from UCI rather than redistributed in the repository.
(6) All new data sets and data sets that are not publicly available are described in detail, including relevant statistics, the data collection process and annotation process if relevant. **Yes.** Appendix C specifies the generative simulation models, and Appendix I gives the evaluation protocols. Section 6.9 and Appendix I.1 describe the recorded measurements, sample counts, collection periods and construction of the sensor-access benchmark. Sensor labels are inherited from the source dataset. No new human annotation is introduced.
(7) All methods used for preprocessing, augmenting, batching or splitting data sets (e.g., in the context of hold-out or cross-validation) are described in detail. **Yes.** Appendix I.1 describes duplicate handling, batch/class-stratified splitting of batches 1–6, training-only standardization and quantization, likelihood estimation, and frozen evaluation on batches 7–10. Exact row assignments, model parameters and source hashes are committed. The study uses the recorded cases without synthetic data augmentation.

Explanations on any of the answers above (optional):

The pinned environment and archived policy outputs permit reconstruction of the reported estimates. Simulation and bootstrap seeds are explicit. Floating-point results and numerical tie decisions can differ across processors and system libraries, so exact reproduction uses the recorded environment or saved trajectories. Timings describe the measured host and are not hardware-independent performance guarantees.